



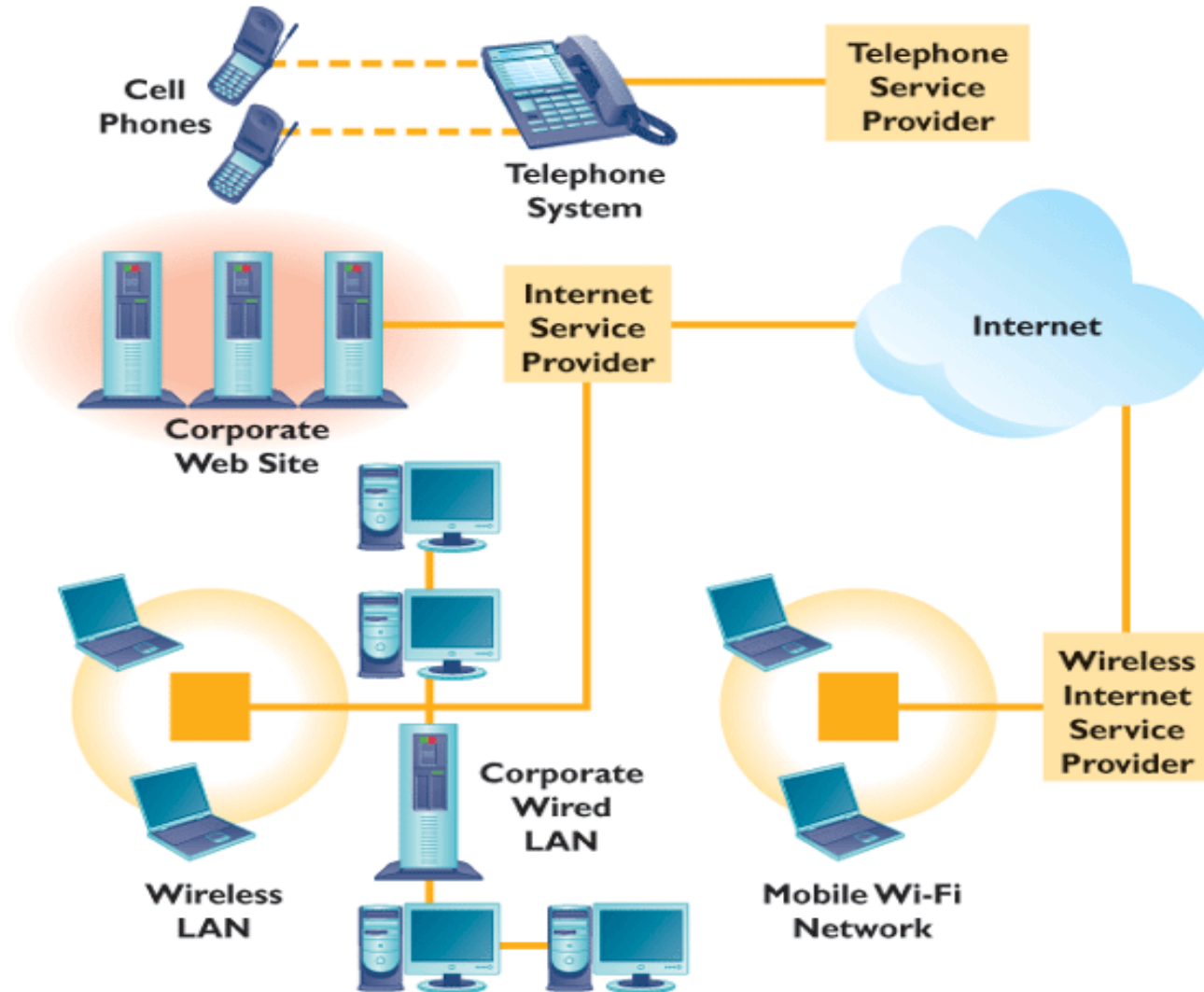
TELECOMMUNICATIONS, NETWORKS, AND WIRELESS COMPUTING



The Corporate Telecommunications System Telecommunications System

- Facilitation of electronic communication
- Telephone systems
- Broadcast and cable TV
- Radio, satellite, and local area networks
- Internet
- Analog or digital

A corporate telecommunications system



Computer Network

Def: A computer network, is a collection of computers and devices connected by communications channels that **facilitates communications** among users and allows users to share resources with other users.

Def2: computer network means an interconnected set of **autonomous computers**.

The term **autonomous** implies that the computers can **function independent of others**.

ADVANTAGES OF NETWORK

- **Speed;** Sharing and transferring files within networks are very rapid. Thus saving time, while maintaining the integrity of the file.
- **Cost;** Individually licensed copies of many popular software programs can be costly. Networkable versions are available at considerable savings. Shared programs, on a network allows for easier **upgrading** of the program on one single file server, instead of upgrading individual workstations.
- **Security;** Sensitive files and programs on a network are passwords protected (established for specific directories to restrict access to authorized users) or designated as "copy inhibit," so that you do not have to worry about illegal copying of programs.

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- **Resource Sharing.** Resources such as, printers, fax machines and modems can be shared.
- **File Sharing:** The major advantage of a computer network is that it allows file sharing and remote file access.

It saves the time which is wasted in copying a file from one system to another, by using a storage device. In addition to that, many people can access or update the information stored in a database, making it up-to-date and accurate.

- **Increased Cost Efficiency:** There are many software available in the market which are costly and take time for installation.
- **Centralized Administration**

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- **Increased Storage Capacity:** As there is more than one computer on a network which can easily share files, the issue of storage capacity gets resolved to a great extent.
- **Flexible Access.** Access their files from computers throughout the firm.
- **Fewer peripherals needed.**
- **Increased communication capabilities.**
- **Avoid File duplication and corruption;** Document tracking software can be used to prevent users from overwriting files, or changing files that others are accessing at the same time.

Disadvantages of Networks

- **Security Issues:** If a computer is a standalone, physical access becomes necessary for any kind of data theft. However, if a computer is on a network, a computer hacker can get unauthorized access by using different tools.
- **Rapid Spread of Computer Viruses:**
- **Expensive Set Up:**
- **Dependency on the Main File Server:** In case the main File Server of a computer network breaks down, the system becomes useless.

Types of Computer Network

Local Area Network

- A local area network (LAN) is a network that connects computers and devices in a **limited geographical area** such as home, school, computer laboratory, office building, or closely positioned group of buildings.
- Each computer or device on the network is a **node**.
- LANs are distinguished from other kinds of networks by three categories: their **size**, **transmission technology** and **topology**.

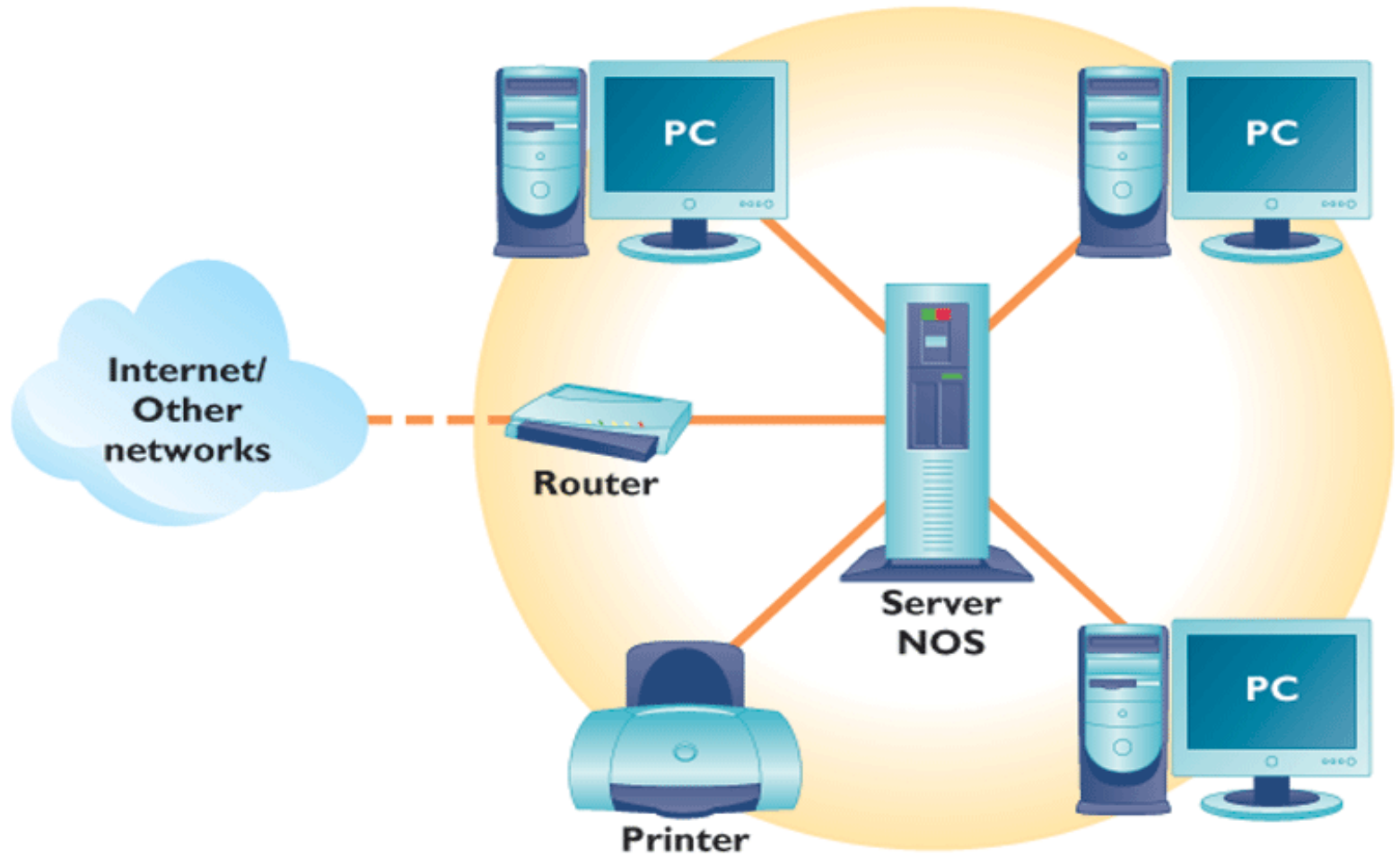
Characteristics of LANs

- Their **higher** data transfer rates,
- Smaller geographic range,
- LAN technologies operate at speeds up to 10 Gbit/s. This is the data transfer rate.
- LANs can be connected to Wide area network by using **routers**.

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- LAN typically used transmission technology consisting of single cable to which all machines are connected.
- Traditional LANs run at speeds of 10 to 100 Mbps
- The most common LAN topologies are **bus, ring and star.**

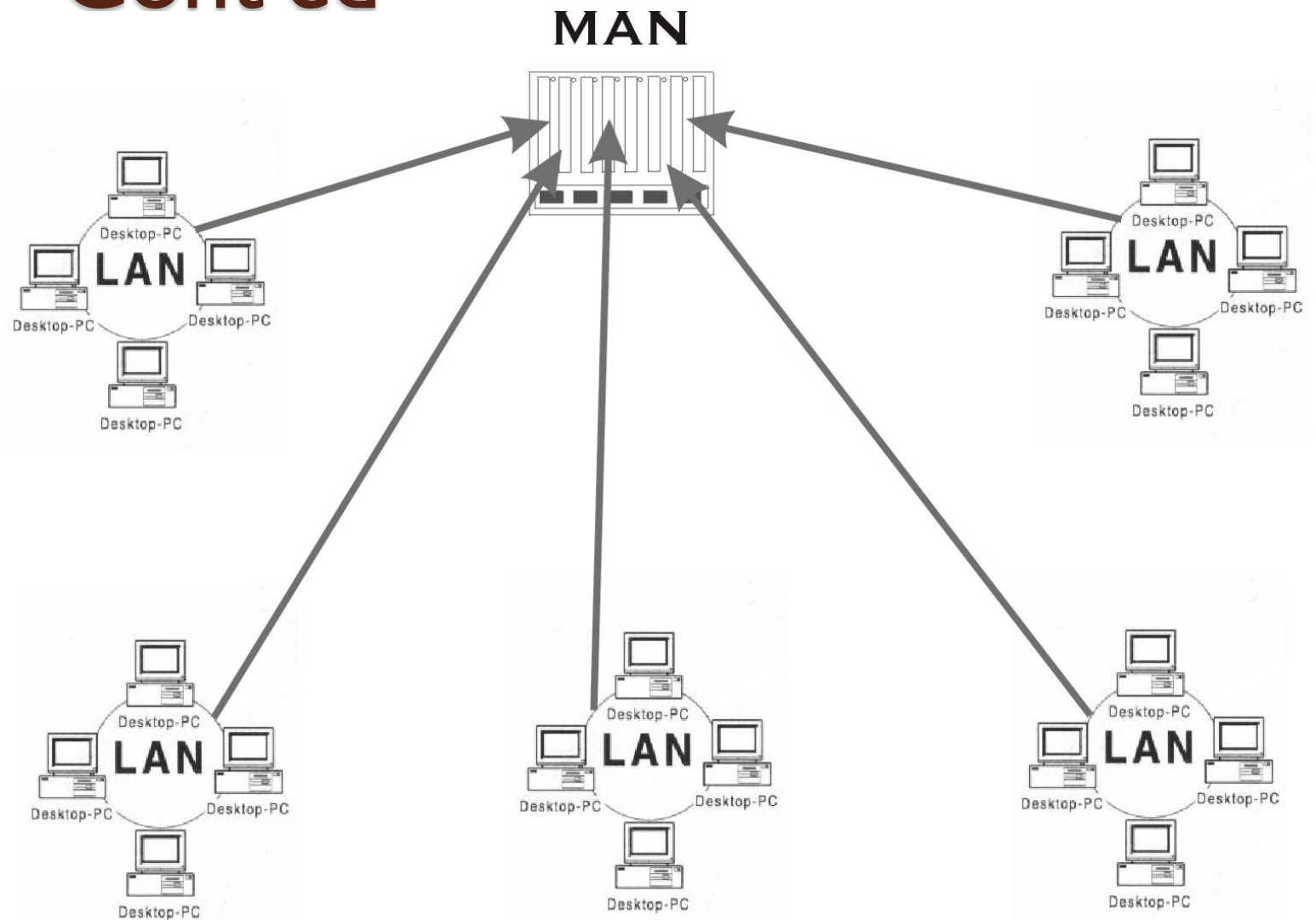
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Metropolitan area network

- MAN is designed to extend over the **entire city**.
- It may be a single network as a **cable TV network** or it may be means of connecting a number of LANs into a larger network so that resources may be shared.
- MAN is wholly owned and operated by a **private company** or may be a service provided by a public company.

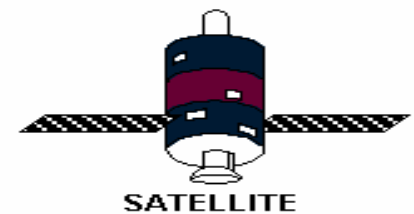
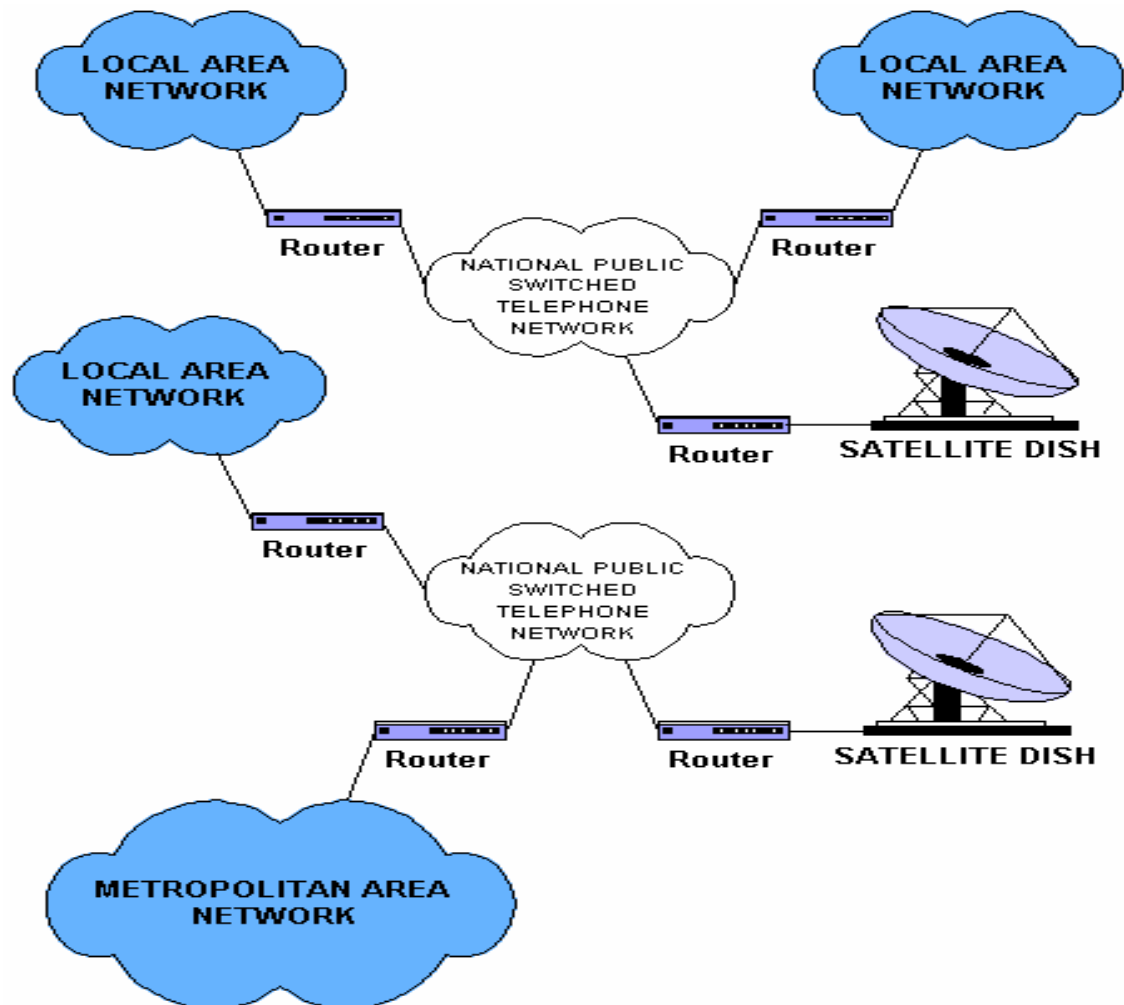
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Wide Area Network (WAN)

- A wide area network (WAN) is a computer network that covers a **large geographic area** such as a country, or spans even intercontinental distances, using a communications channel that combines many types of media such as telephone lines, cables, and air waves.
- A WAN is a **geographically-dispersed collection of LANs**. A network device called a router connects LANs to a WAN.

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Client/server networks

- A client/server network environment is one in which **one computer** acts as the **server** and provides data distribution and security functions to **other computers** that are independently running various applications.
- Windows domain network model, uses a **dedicated server** to **manage the computers** in the network.
- When a PC requests a file, the server searches all active peers on the network.
- The server sends the requesting PC a list of links to all active peers who have the file

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Advantage

- Provides the network with a platform that can better **protect the integrity** and security of the content and users of the network.
- Up gradation and Scalability in Client-server set-up .
- **Accessibility** : From various platforms in the network, server can be accessed remotely.

Disadvantage

- The server can be slowed or overwhelmed by too many users or technical problems.
- You **need professional IT people** to maintain the servers and other technical details of network.

Peer-to-peer networks

- Peer-to-peer networks treats all **processors equally** and is used primarily in small networks with 10 or fewer users.
- The various computers on the network can exchange data by **direct access** and can share peripheral devices without going through a separate server.

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- No central directory or server
- File-sharing software connects one PC to another online user.
- When you request a file, the software searches every online user and sends you a list of active file names.
- Clicking a link automatically transfers the file from that user's hard drive to yours.

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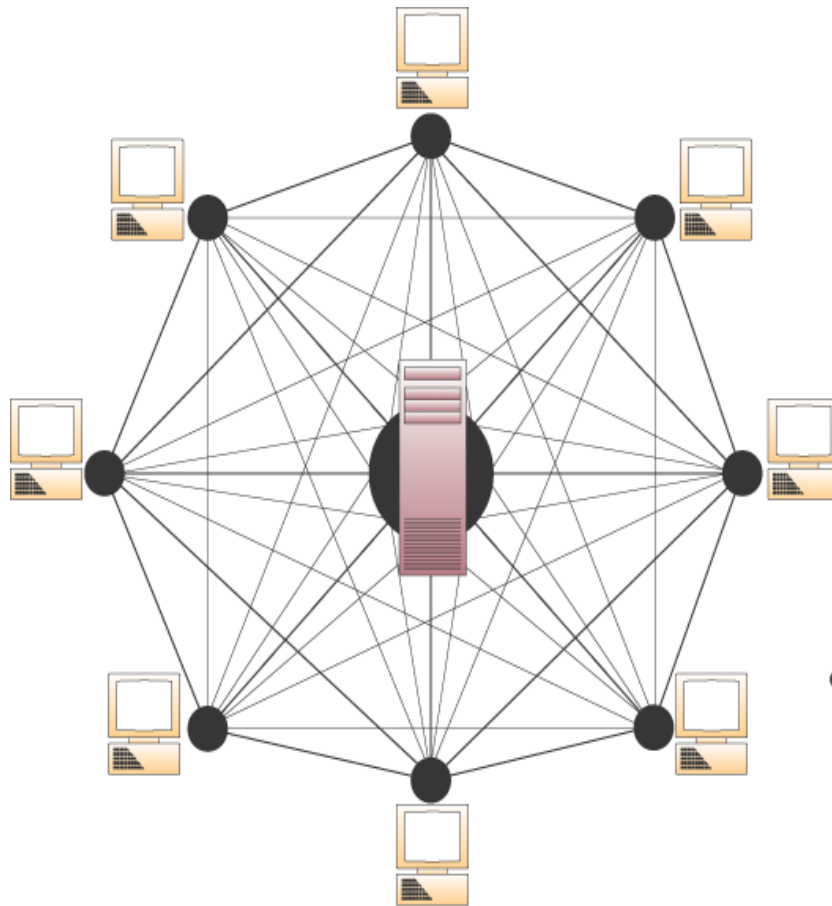
Advantages

- Direct access to data(faster)
- There is no need for full-time System Administrator.
- It is easy to install and so is the configuration of computers on this network.

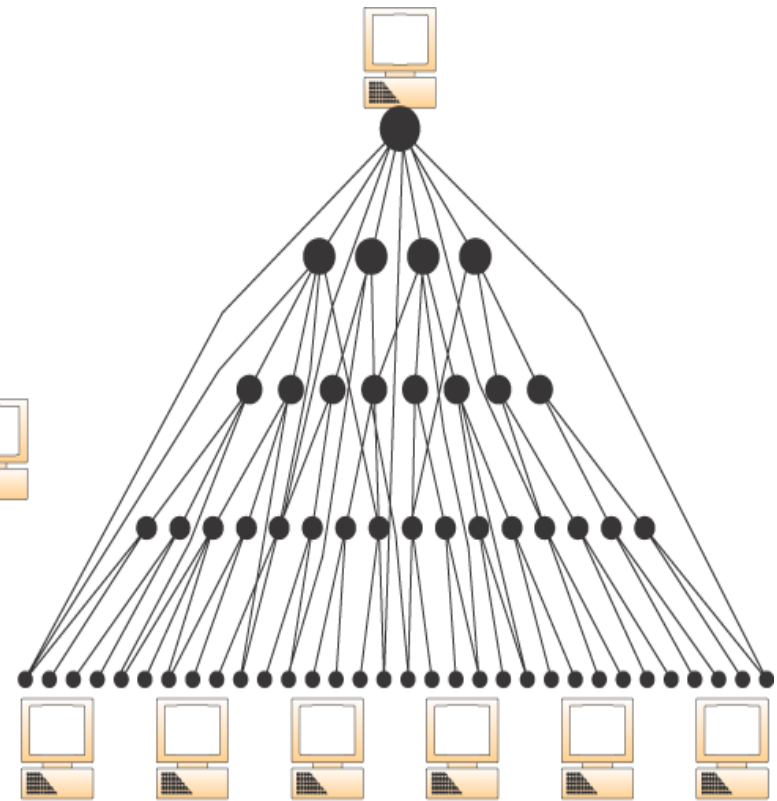
Disadvantages

- The whole system is decentralized thus it is difficult to administer.
- Data recovery or backup is very difficult.
- Security in this system is very less viruses, spywares, trojans, etc malwares can easily transmitted over this P-2-P architecture.

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- A peer-to-peer network architecture with a directory of all peers on a central server

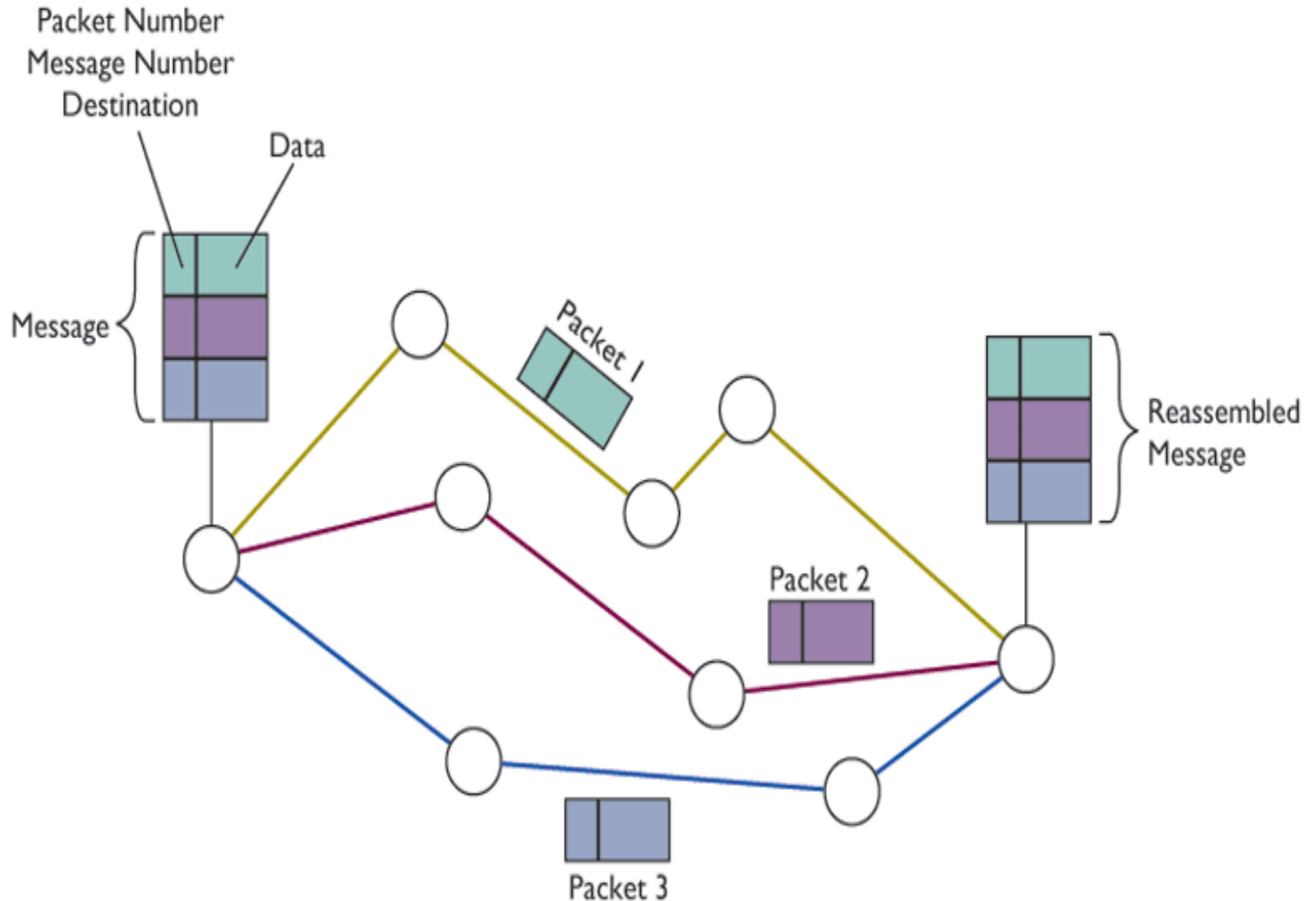


- A pure peer-to-peer network architecture with no central directory server

Packet Switching

- Is a method of **slicing** digital messages into parcels called **packets**, sending the packets along different communication paths as they become available, and then **reassembling** the packets once they arrive at their destinations.
- Messages broken into “**packets**” before transmission
- Packets include information for **directing the packets to the right destination** and **error-checking**.
- Packets travel independently using **routers**; **reassembled** into original message at destination.

Packet-switched networks and packet communications



Network topologies

- **Network topology** is the layout pattern of **interconnections** of the various elements (links, nodes, etc.) of a **computer network**.
- The **way in which the elements** of a network are **mapped or arranged** is known as a **network topology**.
- A topology describes the **physical and the logical** interconnection between the different nodes of a network.

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- **Physical topology** means the physical design of a network including the devices, location and cable installation.
- **Logical topology** refers to **how data** is actually transferred in a network as opposed to its physical design.

Types of Network Topologies

- Network topologies are categorized into three major basic types:

- Bus
- Ring
- Star

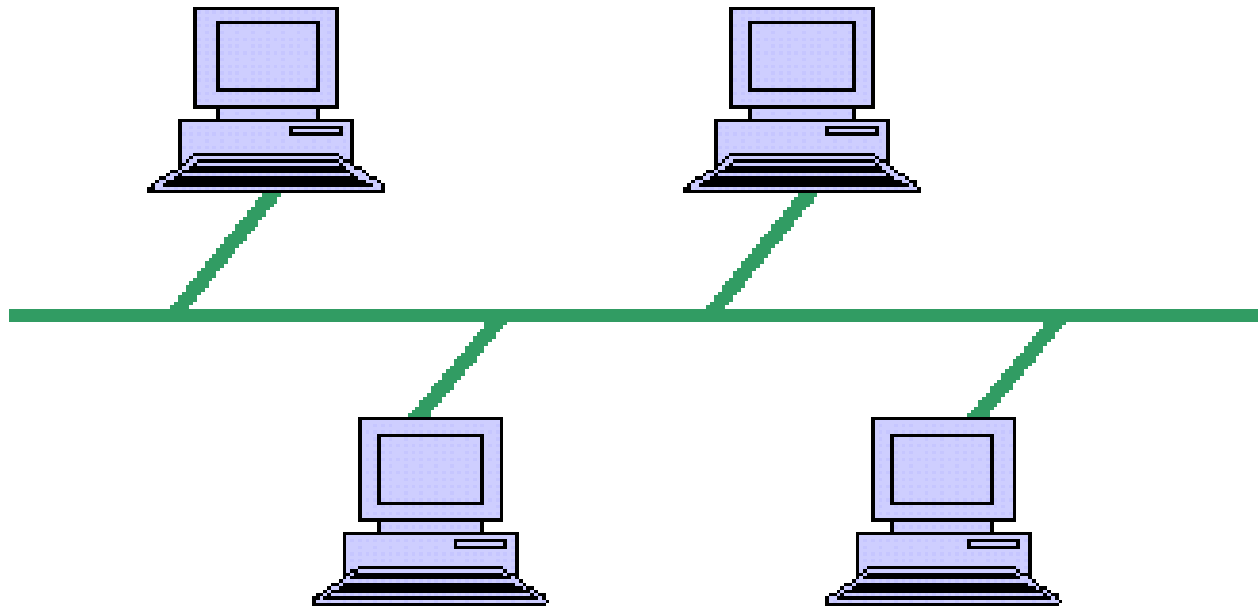
Others include;

- Tree
- Mesh

Bus Topology

- All devices are connected to a **central cable**, called the **bus or backbone**.
- A single cable, the backbone functions as a **shared communication medium** that devices attach or tap into with an interface connector.
- A device wanting to communicate with another device on the network sends a **broadcast message onto the wire** that all other devices see, but **only the intended recipient** actually **accepts and processes** the message.

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Advantages of Bus topology

- Cheap because of its simplicity
- Requires a short cable length
- Easy to expand the network
- Simple to setup compared to Star and Ring topology
- There is no chances of data collision since one computer transmit at a time
- Locating cable fault in a bus topology is relatively easy
- Bus topology is ideal for one to many data transmission

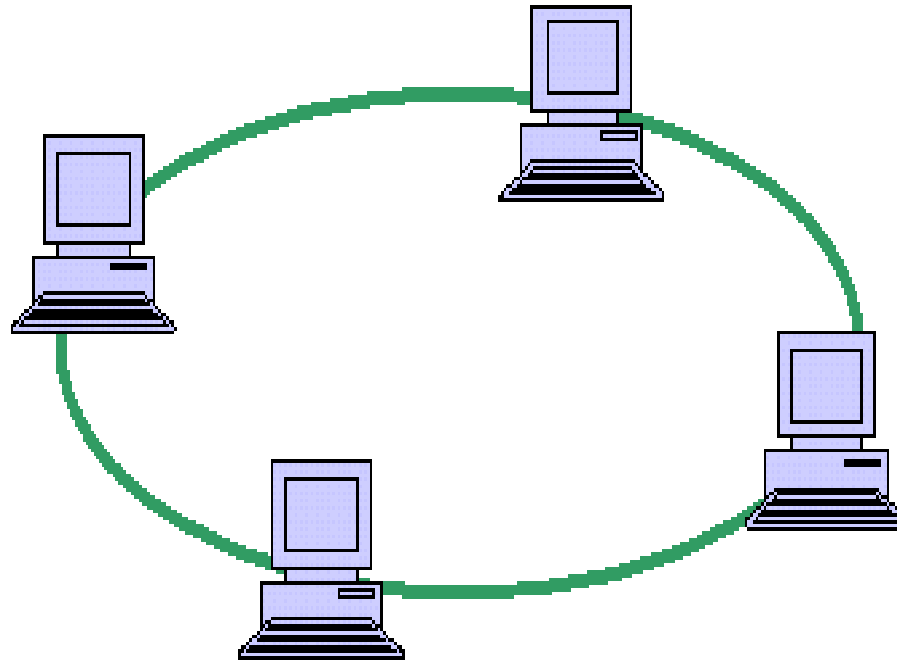
Disadvantages of Bus topology

- **Fault diagnosis** is difficult in a bus topology – detection of fault may have to be performed from any point in the network
- **Fault isolation is difficult** – if fault in the node is detected the node can simply be removed but if the fault is in network medium itself and entire segment of the Bus must be disconnected to isolate the fault
- **Repeater configuration** – When a Bus type network has its backbone intended for using a repeater, configuration maybe necessary thus may involve tailoring cable length, adjusting terminator etc

Ring Topology

- All devices are connected to **one another** in the shape of a closed loop, so that each device is connected directly to **two other devices**, one on either side of it.
- All messages travel through a ring in the same direction (either "clockwise" or "counterclockwise").
- The ring topology does not require a **central server** to manage connectivity between the nodes and facilitates an orderly network operation.
- A failure in any cable or device breaks the loop and can take down the entire network.

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Advantages and Disadvantages of Ring Topology

Advantages

- The ability to achieve transmission rates of the order of 10 million bits per second.
- Provision of local communication via a single channel.
- No central server which reduces the cost of establishing the ring topology.

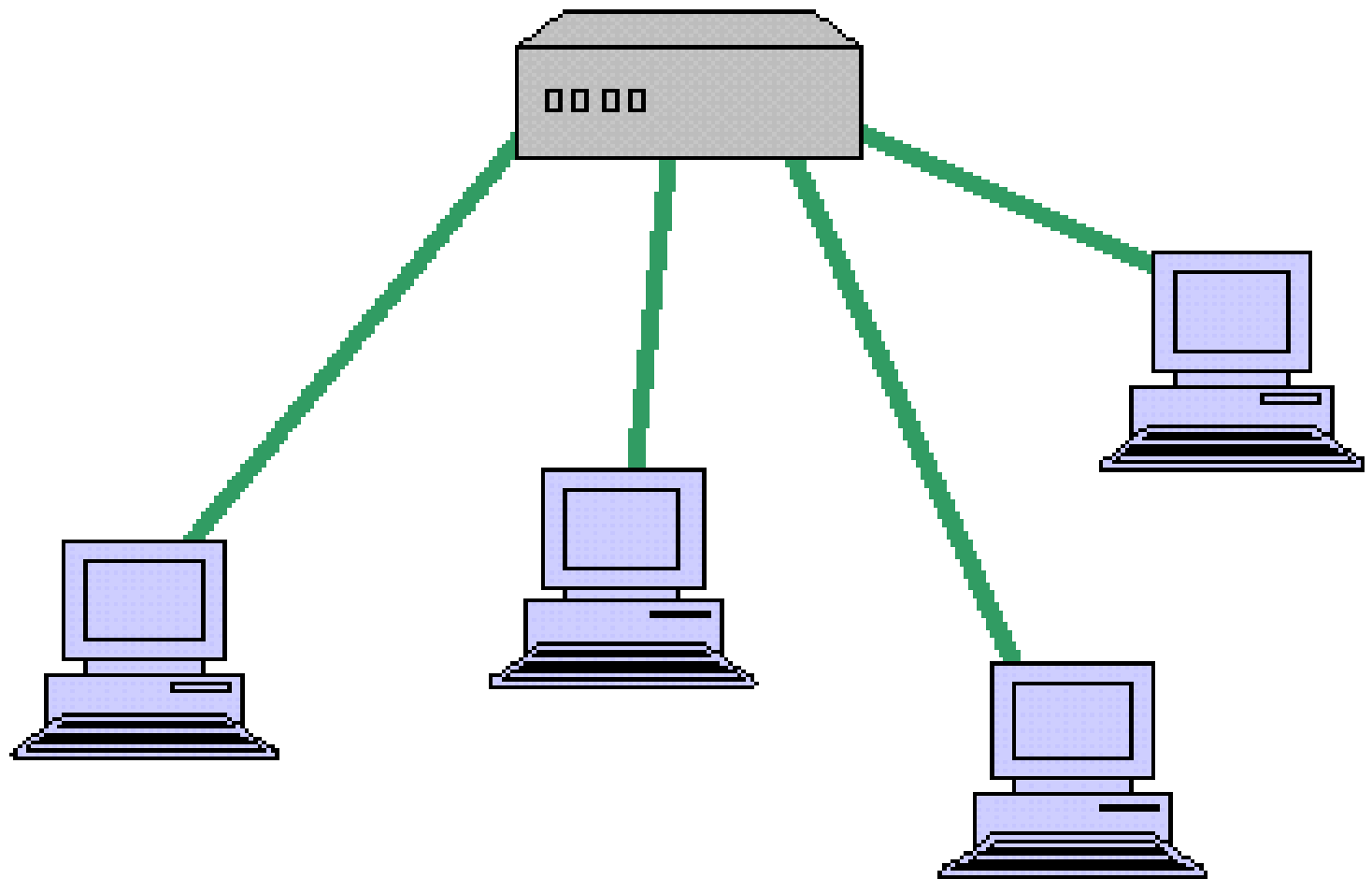
Disadvantages

- Failure of one node result into entire network failure.
- Detection of fault is very difficult in ring topology.
- Isolation of fault is not easy in this topology.

Star Topology

- Each node of the network is connected to a **central node**, which is known as a **hub**. The data that is transmitted between the network nodes passes across the **central hub**.
- Devices typically connect to the hub with Unshielded Twisted Pair (UTP) Ethernet.
- Nodes communicate across the network by passing data through the hub.
- Compared to the bus topology, a star network generally requires **more cable**, but a failure in any star network cable will only take down **one computer's network** access and **not the entire LAN**.

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Advantages of star topology

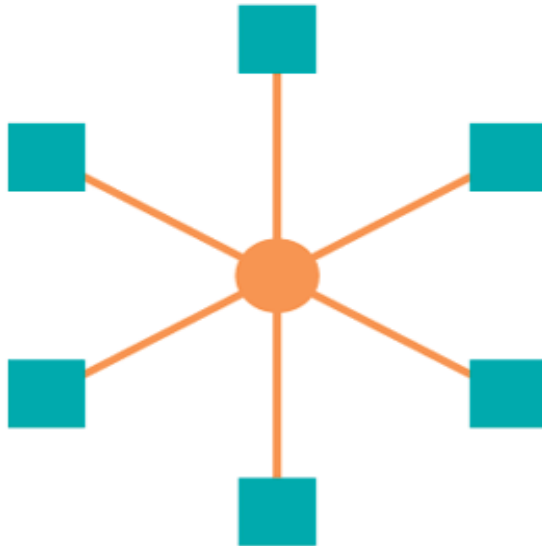
- Diagnostic and isolation of problem is easy
- Easy to add a new computer system to the network
- Failure of one workstation does not affect the entire network
- Uses a single access protocols
- Provide ease of service
- Is very fast

Disadvantages of star topology

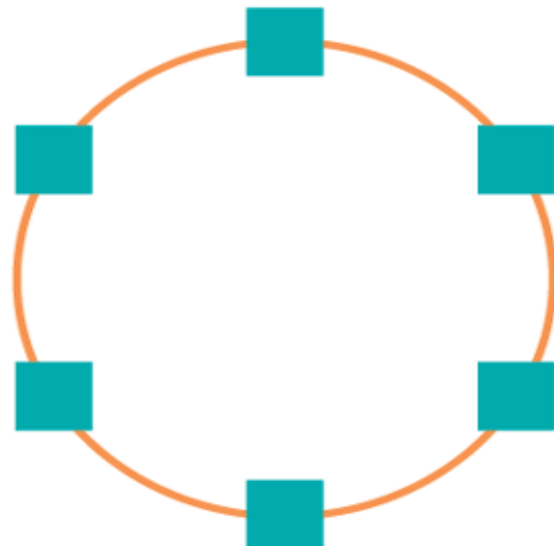
- There is central node dependency. The failure of the central hub results renders the entire network inoperable.
- Becomes expensive when purchasing the nodes.
- Always requires very long cable length when connecting many nodes.

summary

Star topology



Ring topology



Bus topology



Other..topologies

Tree Topology

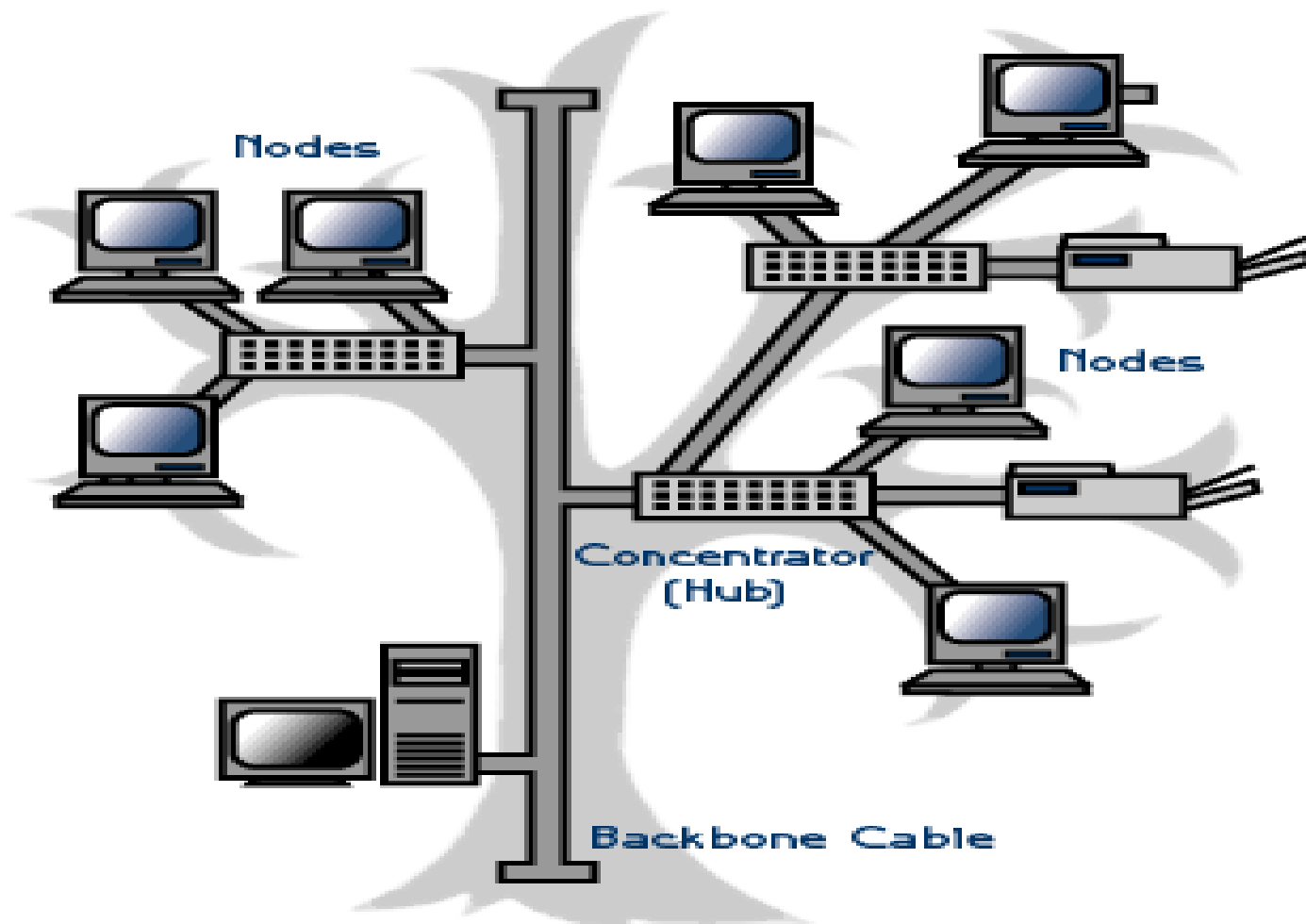
- It is also known as a **hierarchical topology** and has a **central root node** that is connected to one or more nodes of a lower hierarchy.
- In a **symmetrical hierarchy**, each node in the network has a specific fixed number of nodes connected to those at a lower level.

Advantages of a Tree Topology

Advantages

- Point-to-point wiring for individual segments.
- Supported by several hardware and software vendors.

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Mesh Topology

- In a full mesh network, **each network node is connected to every other node** in the network. Due to this arrangement of nodes, it becomes possible for a simultaneous transmission of signals from one node to several other nodes.
- In a partially connected mesh network, **only some of the network nodes are connected to more than one node**. This is beneficial over a fully connected mesh in terms of redundancy caused by the point-to-point links between all the nodes.

Advantages and Disadvantage of Mesh Topology

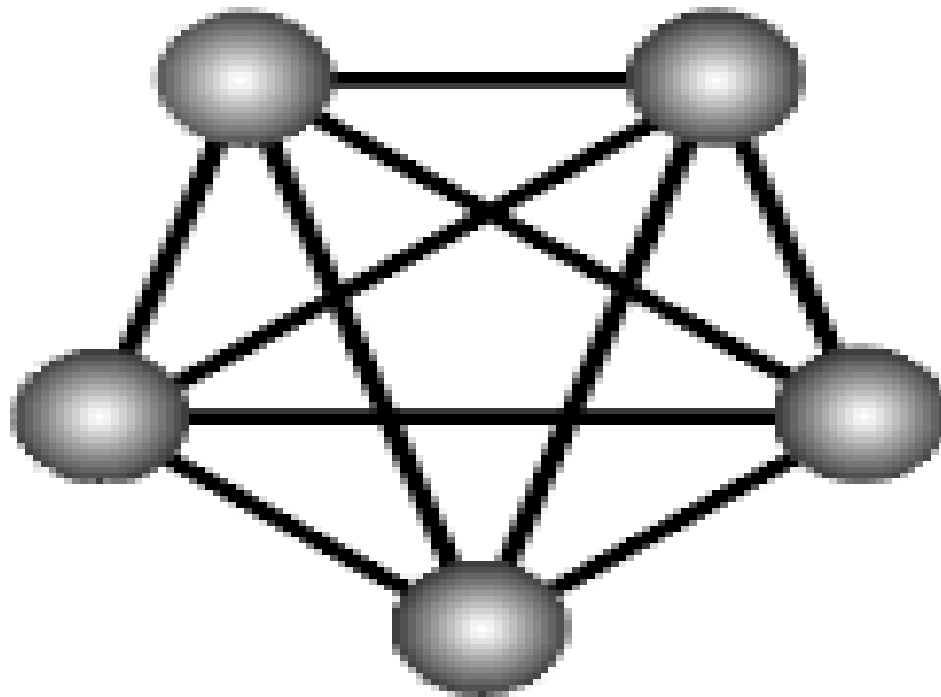
Advantage

The arrangement of the network nodes is such that it is possible to transmit data from one node to many other nodes at the same time.

Disadvantage

The arrangement wherein every network node is connected to every other node of the network, many of the connections serve no major purpose. This leads to the redundancy of many of the network connections.

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Mesh network

TCP/IP

TCP/IP

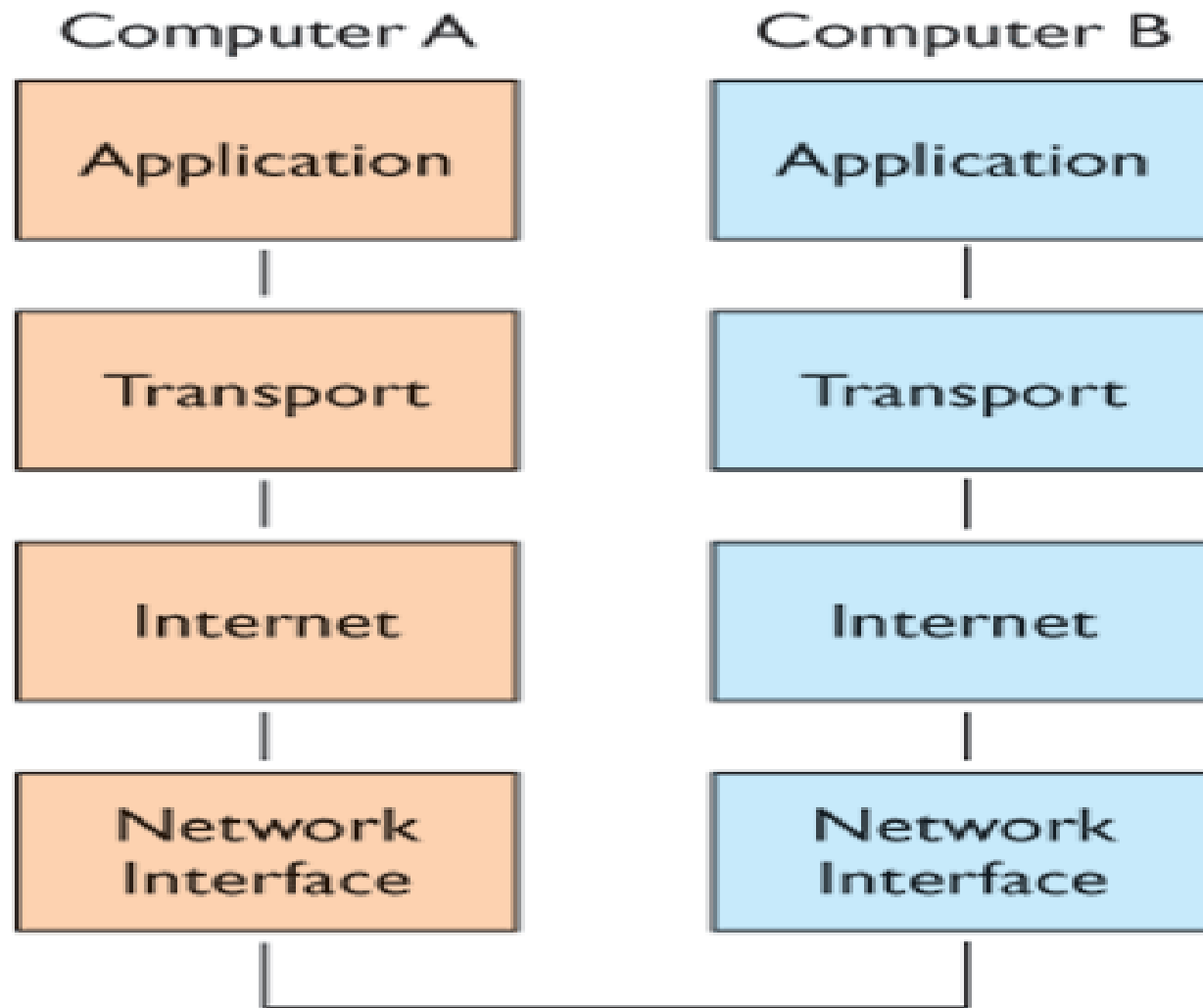
Transmission Control Protocol/Internet Protocol

- Open suite of protocols for connectivity developed in 1970s
- Provides standards for breaking messages into packets, **routing** them to destination addresses, and **reassembling** them at end.
- Allows for **communication** regardless of hardware/software

TCP/IP: Four-Layer Reference Model

1. **Application layer:** Communication between applications and other layers.
2. **Transport layer:** Acknowledging and sequencing packets to/from application.
3. **Internet layer:** Addressing, routing, packaging data packets.
4. **Network interface layer:** Placing packets on and receiving them from network medium.

The Transmission Control Protocol/Internet Protocol (TCP/IP) reference model



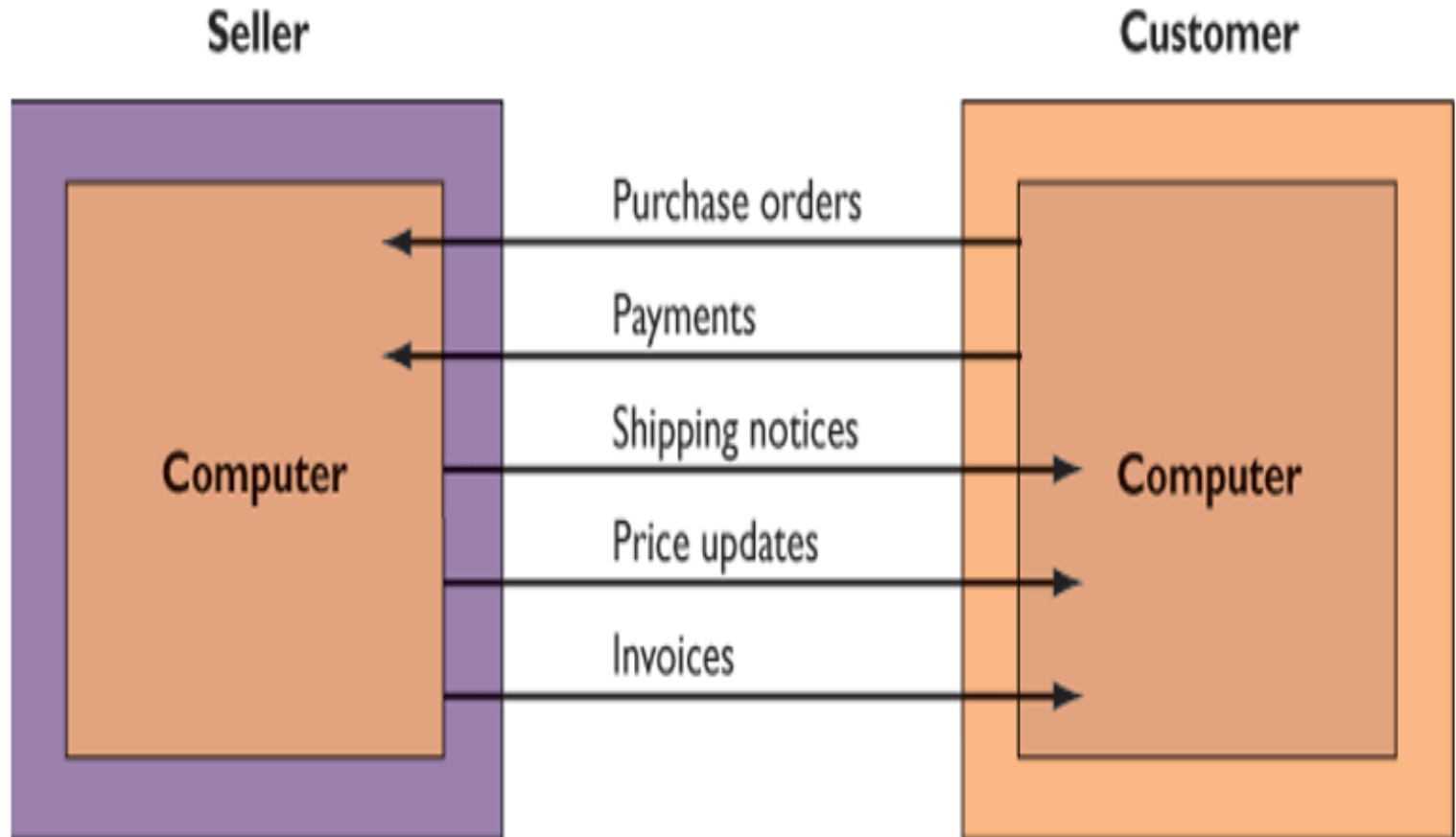
Teleconferencing, Dataconferencing, and Videoconferencing

- **Teleconferencing**
 - Ability to confer with a group of people simultaneously
- **Data conferencing**
 - Two or more users can edit and modify data files simultaneously
- **Videoconferencing**
 - Participants are able to see each other over video screens

Electronic Data Interchange

- Computer-to-computer exchange between two organizations of standard transaction documents, such as invoices, purchase orders.
- Minimizes paper-handling and data input; lowers transaction costs.
- Transmits structured data with fields, unlike e-mail.

Electronic data interchange (EDI)



Internet Services and Communication Tools

- E-mail
- Electronic discussion groups
- Chatting
- Instant messaging
- Telnet
- File transfer protocol
- WWW(World Wide Web)